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Catalog Integration

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1

Enroll in Private Connectivity Feature

Contact Snowflake account team to enable private connectivity for Azure environments.

2

Provision Azure Private Endpoint

Create Private Endpoint in Snowflake VNet to connect to Unity Catalog via Azure Private Link.

- Execute: `SYSTEM$PROVISION_PRIVATELINK_ENDPOINT('<azure_private_link_service_id>', '<databricks_workspace_URL>')`
- Obtain Private Link service ID from Databricks Azure documentation
- For multiple workspaces in same region, use wildcard pattern in workspace URL
- Monitor endpoint provisioning: `SYSTEM$GET_PRIVATELINK_ENDPOINTS_INFO()`
- Confirm status transitions from 'pending' to 'available'

3

Register Snowflake Private Endpoint in Azure Databricks

Configure Databricks to accept connections from Snowflake Private Endpoint.

- Query `SYSTEM$GET_PRIVATELINK_ENDPOINTS_INFO()` and extract 'snowflake_endpoint_name'
- In Databricks Azure console: Register the Private Endpoint connection
- Approve the Private Endpoint connection in Databricks
- Create Private Access Settings in Databricks and link to approved endpoint
- Verify Private Link Service configuration in Azure portal

4

Create Catalog Integration with Private Connectivity

Configure Unity Catalog integration using Azure Private Link.

- `CREATE CATALOG INTEGRATION` with Azure-specific configuration
- Set `CATALOG_SOURCE = ICEBERG_REST`
- Set `TABLE_FORMAT = ICEBERG`
- IMPORTANT: Specify `CATALOG_API_TYPE = PRIVATE` in `REST_CONFIG`
- Configure authentication (`BEARER_TOKEN` or `OAuth`)
- Point `CATALOG_URI` to Azure Databricks Unity Catalog REST endpoint
- Enable the integration

5

Configure External Volume for Azure Storage

Set up private connectivity to Azure Data Lake Storage Gen2 or Blob Storage.

- Create external volume with Azure Private Endpoint configuration
- Configure Azure Private Endpoint for storage account
- Ensure Snowflake can write data via Private Link
- Enable Unity metadata access through Azure Private Link
- Configure managed identity or service principal for authentication
- Verify network isolation and data path security

Connectivity Strategy

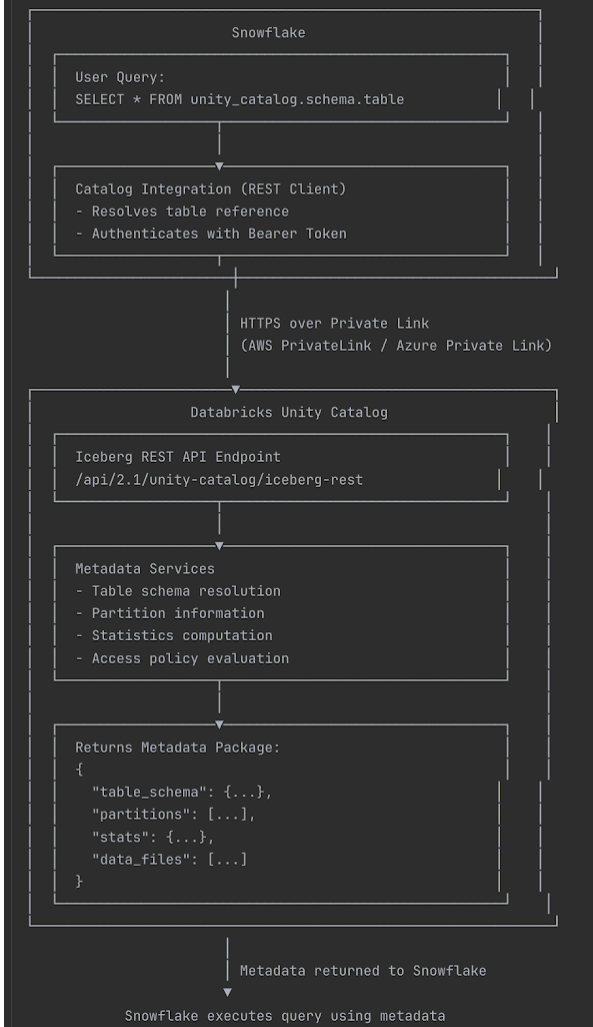
Option 1: Wildcard Workspace URL (Recommended)

- ✓ Use wildcard pattern when provisioning endpoint: '*.databricks.com' or '*.azuredatabricks.net'
- ✓ Single private endpoint serves all workspaces in the region
- ✓ Simplifies management and reduces overhead

Option 2: Individual Endpoint Provisioning

- Provision separate endpoint for each workspace URL
- Provides workspace-level isolation if required
- More complex to manage but offers granular control

Metadata Extraction Process



Problem:

Rewrite the problem statement as to find the most efficient way to extracting metadata from Databricks to Snowflake. **Solution:**

Solution Architecture - Shows how Unity Catalog integration solves these through:

- Direct metadata extraction via Iceberg REST API
- Real-time metadata synchronization
- Query-time metadata resolution (no data replication needed)
- Unified data discovery experience